

Problem 1. Lines in $\mathbb{R}^3$.

- (a) Find a parametrization for the line passing through $P = (1, 0, 0)$ and $Q = (1, 2, 3)$.
- (b) Find a parametrization for the line of intersection of the following two planes:

$$\begin{cases} x + 0 + -z = 2, \\ 0 - y + 3z = 1. \end{cases}$$

Problem 2. Integration of Vectors. Consider a curve $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ satisfying

$$\mathbf{r}(0) = \langle 0, 0 \rangle,$$

$$\mathbf{v}(0) = \mathbf{r}'(0) = \langle 1, 1 \rangle,$$

$$\mathbf{r}''(t) = \mathbf{v}'(t) = \mathbf{a}(t) = \langle 0, t \rangle.$$

[Note: The acceleration is not constant.]

- (a) Integrate $\mathbf{a}(t)$ to obtain $\mathbf{v}(t)$.
- (b) Then integrate $\mathbf{v}(t)$ to obtain $\mathbf{r}(t)$.